Special Relativity Notes

EP-2108 Mid-Semester Examination — Solutions

13/10/26

Question 1

An event occurs in frame S at a point A. A second event occurs at point B in the same frame $0.01\,\mu s$ later. The distance between A and B as measured by an observer in frame S is 5 meters. As viewed by an observer in another frame S', the two events occur simultaneously but at a distance of 4 meters apart. Calculate the velocity of the frame S' with respect to S. If possible, calculate the velocity of a frame S'' where the two events occur at the same place. Give full justification for your answer.

Solution

In frame S: $\Delta t = 0.01\,\mu s = 1 \times 10^{-8}\,\text{s}$, $\quad \Delta x = 5\,\text{m}$.

Velocity of S'

S' sees the events as simultaneous: $\Delta t' = 0$. Using the Lorentz transformation for the time coordinate,

$$\Delta t' = \gamma\left(\Delta t - \frac{v\,\Delta x}{c^2}\right)$$

Setting $\Delta t' = 0$:

$$\Delta t = \frac{v\,\Delta x}{c^2} \quad \Longrightarrow \quad \boxed{v = \frac{c^2 \Delta t}{\Delta x}}$$

$$v = \frac{(3\times10^8)^2 \times 10^{-8}}{5} = \frac{9\times10^{16}\times10^{-8}}{5} = \frac{9\times10^8}{5} = 1.8\times10^8\,\text{m/s}$$

$$\boxed{v = 1.8\times10^8\,\text{m/s} = 0.6\,c} \qquad \text{(directed from } A \text{ to } B\text{)}$$

Consistency check ($\gamma = 1/\sqrt{1-0.36} = 1/0.8 = 1.25$):

$$\Delta x' = \gamma(\Delta x - v\Delta t) = 1.25\,(5 - 1.8\times10^8\times10^{-8}) = 1.25\,(5-1.8) = 1.25\times3.2 = 4\,\text{m}$$

This matches the given $\Delta x' = 4\,\text{m}$ exactly, confirming the value of v.

Can a frame S'' exist where the two events occur at the same place?

We would need $\Delta x'' = 0$:

$$\Delta x'' = \gamma''(\Delta x - v''\Delta t) = 0 \quad \Longrightarrow \quad v'' = \frac{\Delta x}{\Delta t} = \frac{5}{10^{-8}} = 5 \times 10^8\,\mathrm{m/s}$$

Since $v'' = 5 \times 10^8\,\mathrm{m/s} > c$, this is **not physically possible**. No such frame S'' exists.

Justification via the invariant interval. With $c\,\Delta t = 3 \times 10^8 \times 10^{-8} = 3\,\mathrm{m}$,

$$\Delta s^2 = c^2\Delta t^2 - \Delta x^2 = (3)^2 - (5)^2 = 9 - 25 = -16\,\mathrm{m}^2 < 0$$

The interval between the two events is **space-like** ($\Delta x > c\,\Delta t$). A space-like interval is invariant, and only a *time-like* interval ($c\Delta t > \Delta x$) can be reduced to zero spatial separation by a boost with $v < c$. Since the interval here is space-like, it can instead be reduced to zero *time* separation (which is exactly what S' achieves), but never to zero *space* separation by any physical ($v < c$) frame.

Question 2

Let us consider an inertial frame S' that moves with a velocity $\vec{v} = \beta c(\cos\phi\,\hat{i} + \sin\phi\,\hat{j})$ with respect to frame S. If the origins of the two frames coincide at $t = t' = 0$ and their axes are parallel to each other, then find the Lorentz matrix Λ for the transformation $(x^0, x, y, z) \to (x'^0, x', y', z')$, where $x^0 = ict$.

Solution

The boost direction is $\hat{n} = \cos\phi\,\hat{i} + \sin\phi\,\hat{j}$, so the components of $\vec{\beta}$ are

$$\beta_1 = \beta\cos\phi, \qquad \beta_2 = \beta\sin\phi, \qquad \beta_3 = 0, \qquad |\vec{\beta}|^2 = \beta^2.$$

For a general boost with $x^0 = ict$, the Lorentz matrix is (derived earlier by splitting $\vec{x}$ into components parallel and perpendicular to $\vec{v}$):

$$\Lambda^0{}_0 = \gamma, \qquad \Lambda^0{}_i = -i\gamma\beta_i, \qquad \Lambda^i{}_0 = i\gamma\beta_i, \qquad \Lambda^i{}_j = \delta_{ij} + (\gamma - 1)\frac{\beta_i\beta_j}{\beta^2}$$

Substituting $\beta_1 = \beta\cos\phi$, $\beta_2 = \beta\sin\phi$, $\beta_3 = 0$:

$$\Lambda^1{}_1 = 1 + (\gamma - 1)\cos^2\phi, \qquad \Lambda^2{}_2 = 1 + (\gamma - 1)\sin^2\phi, \qquad \Lambda^3{}_3 = 1$$
$$\Lambda^1{}_2 = \Lambda^2{}_1 = (\gamma - 1)\sin\phi\cos\phi, \qquad \Lambda^1{}_3 = \Lambda^3{}_1 = 0, \qquad \Lambda^2{}_3 = \Lambda^3{}_2 = 0$$

$$\Lambda = \begin{pmatrix} \gamma & -i\gamma\beta\cos\phi & -i\gamma\beta\sin\phi & 0 \\ i\gamma\beta\cos\phi & 1 + (\gamma - 1)\cos^2\phi & (\gamma - 1)\sin\phi\cos\phi & 0 \\ i\gamma\beta\sin\phi & (\gamma - 1)\sin\phi\cos\phi & 1 + (\gamma - 1)\sin^2\phi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Check: setting $\phi = 0$ (boost purely along x) reduces this to the familiar matrix $\begin{pmatrix} \gamma & -i\beta\gamma & 0 & 0 \\ i\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$, as expected.

Question 3

A ρ-meson of mass $760\,\mathrm{MeV}/c^2$ decays at rest into two π-mesons of mass $140\,\mathrm{MeV}/c^2$ each. What is the relative velocity of the π-mesons with respect to each other?

Solution

Energy of each pion. The ρ-meson decays at rest, so total momentum is zero. By symmetry the two pions (equal mass) fly off back-to-back with equal and opposite momenta, hence equal energies:

$$E_\pi = \frac{M_\rho c^2}{2} = \frac{760}{2} = 380\,\mathrm{MeV}$$

Momentum of each pion (from $E^2 = p^2c^2 + m^2c^4$):

$$(pc)^2 = E_\pi^2 - (m_\pi c^2)^2 = 380^2 - 140^2 = (380-140)(380+140) = 240 \times 520 = 124800$$

$$pc = \sqrt{124800} = 20\sqrt{312} \approx 353.27\,\mathrm{MeV}$$

Speed of each pion (in the ρ rest frame):

$$\gamma_\pi = \frac{E_\pi}{m_\pi c^2} = \frac{380}{140} = \frac{19}{7} \approx 2.7143$$

$$\beta_\pi = \frac{pc}{E_\pi} = \sqrt{1 - \frac{1}{\gamma_\pi^2}} = \sqrt{1 - \left(\frac{7}{19}\right)^2} = \sqrt{\frac{312}{361}} = \frac{2\sqrt{78}}{19} \approx 0.9297$$

Relative velocity. The two pions move with $+\beta_\pi c$ and $-\beta_\pi c$ along the same line (in the ρ rest frame). Using the relativistic velocity-addition rule to find the speed of one pion as seen from the other's rest frame:

$$v_{\mathrm{rel}} = \frac{\beta_\pi c - (-\beta_\pi c)}{1 - \dfrac{(\beta_\pi c)(-\beta_\pi c)}{c^2}} = \frac{2\beta_\pi c}{1 + \beta_\pi^2}$$

$$\frac{v_{\mathrm{rel}}}{c} = \frac{2\beta_\pi}{1+\beta_\pi^2} = \frac{76\sqrt{78}}{673} \approx 0.9973$$

$$\boxed{v_{\mathrm{rel}} \approx 0.9973\,c \approx 2.992 \times 10^8\,\mathrm{m/s}}$$

Question 4

A particle of rest mass m_0 and speed v collides and sticks to a stationary particle of mass M. What is the final speed of the composite particle?

Solution

Let $\gamma = \dfrac{1}{\sqrt{1 - v^2/c^2}}$. Before the collision:

$$E_i = \gamma m_0 c^2 + Mc^2, \qquad p_i = \gamma m_0 v$$

(the stationary particle contributes zero momentum and rest energy Mc^2).

By conservation of energy and momentum, the composite particle (rest mass M', speed V) has

$$E_f = E_i, \qquad p_f = p_i$$

The speed of any particle is related to its energy and momentum by $V = \dfrac{p_f c^2}{E_f}$:

$$V = \frac{p_i c^2}{E_i} = \frac{\gamma m_0 v\, c^2}{\gamma m_0 c^2 + Mc^2} = \frac{\gamma m_0 v}{\gamma m_0 + M}$$

Multiplying numerator and denominator by $\sqrt{1 - v^2/c^2}$ (i.e. by $1/\gamma$):

$$\boxed{V = \frac{m_0 v}{m_0 + M\sqrt{1 - v^2/c^2}}}$$

Checks: if $M \to 0$, $V \to v$ (no collision partner, unchanged); if $v \to 0$, $V \to 0$ (nothing moving to begin with); if $m_0 \gg M$ or $v \ll c$ so that $\gamma \approx 1$, $V \approx \dfrac{m_0 v}{m_0 + M}$, the familiar non-relativistic inelastic-collision result.

Question 5

A circular lamina moves with its plane parallel to the X-Y plane of a reference frame S at rest. Assuming its motion to be along the X-axis, calculate the velocity at which its surface area would appear to be reduced to half to an observer in frame S.

Solution

Let the lamina have rest radius R_0, so its rest area is $A_0 = \pi R_0^2$.

Length contraction acts **only along the direction of motion** (the X-axis); the Y-direction (perpendicular to the motion) is unaffected. So the circle of radius R_0 becomes an **ellipse**:

- semi-axis along Y: unchanged, $= R_0$
- semi-axis along X: contracted, $= R_0\sqrt{1 - v^2/c^2}$

The area of an ellipse with semi-axes a, b is πab:

$$A = \pi\,(R_0)\left(R_0\sqrt{1 - v^2/c^2}\right) = \pi R_0^2\sqrt{1 - v^2/c^2} = A_0\sqrt{1 - v^2/c^2}$$

Setting $A = \dfrac{A_0}{2}$:

$$A_0\sqrt{1 - v^2/c^2} = \frac{A_0}{2} \quad\Longrightarrow\quad \sqrt{1 - v^2/c^2} = \frac{1}{2} \quad\Longrightarrow\quad 1 - \frac{v^2}{c^2} = \frac{1}{4} \quad\Longrightarrow\quad \frac{v^2}{c^2} = \frac{3}{4}$$

$$\boxed{v = \frac{\sqrt{3}}{2}\,c \approx 0.866\,c \approx 2.598 \times 10^8\ \text{m/s}}$$